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Improving GDP Nowcasting Using Fiscal Variables: Evidence from a Dynamic Factor Model

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Abstract

While traditional literature tends to exclude fiscal data from GDP nowcasting, this study argues that standard models fail to fully capture the growth impulse generated by unprecedented public budget expansions. To address this gap, the work compares an adapted specification of the New York Fed Staff Nowcast against an augmented model that incorporates three federal fiscal indicators.

The analysis, conducted on a reconstruction of real-time vintages from 2017 to 2025, highlights a systematic improvement in accuracy metrics for the fiscal-enhanced model. Specifically, the fiscal signal becomes predominant in two distinct circumstances: during the second-quarter tax season and throughout the post-pandemic period. This work contributes to the literature by demonstrating the necessity of adapting DFM nowcasting models to the new centrality of fiscal policy.

Keywords: nowcast, GDP nowcasting, macroeconomic data, Kalman filter, Dynamic Factor Model, fiscal policy

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1. Introduction

1.1. Context and Motivation

The accurate and timely estimation of Gross Domestic Product (GDP) has long been a cornerstone of economic analysis, guiding decision-making by policymakers, central banks, and financial institutions. Historically, GDP data has been released on a quarterly basis, with significant delays, often creating challenges in responding to real-time economic developments. In an era of rapid economic shifts and increased market volatility, the need for more timely and accurate economic indicators has never been greater. This is particularly true when considering the critical role that GDP plays in shaping policy decisions, influencing investor expectations, and guiding economic interventions.

To address this need, real-time monitoring of macroeconomic conditions has become essential for economists at central banks, government agencies, and private institutions. These professionals sift through vast amounts of data, extracting key information to provide timely economic insights. GDP nowcasting refers to the practice of using high-frequency, real-time data to estimate economic output before official statistics are released, which often have a significant lag. It combines various data sources into models to provide timelier insights into the current economic state for policymakers and businesses.

A key feature of this nowcasting process is its ability to quickly adjust forecasts as new data is released, offering an up-to-date picture of economic growth. In times of high economic uncertainty, nowcasting has become an essential tool for making rapid, informed decisions.

1.2. Research Gap & Thesis Objectives

GDP nowcasting models typically rely solely on macroeconomic data, as GDP is considered a macroeconomic measure. Nevertheless, this thesis is built on the intuition that incorporating

fiscal variables into nowcasting models could reduce the average forecasting error for two main reasons.

The first reason is found in the traditional calculation of GDP, expressed as:

$$Y = C + I + G + NX \quad (1)$$

where C stands for consumption spending, I represents business investments, G denotes government spending, and NX refers to net exports. By not incorporating fiscal variables, nowcasting models may fail to fully capture the portion of output that stems from government spending (G).

The second reason is rooted in recent years, where government budget expansions have led public spending to actively influence economic growth. The explosion in public debt issuance, shown in Appendix Figure A1, has contributed to creating liquidity that supports consumption and investment by households and businesses. Although this effect is reflected in macroeconomic data related to consumption and investment, the transmission of these effects occurs with a delay and may not be entirely captured by these data.

To address this gap, this thesis incorporates fiscal variables into a nowcasting model and assesses the extent to which they contribute to minimize the nowcast error. In doing so, this work contributes to the existing literature on economic nowcasting in three distinct ways.

First, it extends the canonical New York Fed Staff Nowcast framework by structurally integrating fiscal indicators, which are typically absent from standard dynamic factor models, evaluating their role within one of the primary models used by the Federal Reserve for monitoring the U.S. economy.

Second, it highlights the seasonality of fiscal signals, showing that their predictive value peaks systematically in the second quarter due to the cyclical nature of tax receipts.

Finally, it demonstrates that the contribution of fiscal variables is significantly more pronounced in the post-pandemic regime, providing evidence that fiscal policy has become a critical driver of economic fluctuations in the current macroeconomic landscape.

2. Literature Review

2.1. The Origins of Nowcasting

The term "nowcasting" was first coined in meteorology, where it refers to the prediction of present or near-future weather conditions using real-time data that is constantly updated. Economists adopted this terminology to address the limitations posed by the significant publication lags of key indicators, such as GDP, inflation or employment. Economic nowcasting thus refers to the real-time assessment of the current state of the economy through the continual integration of newly released, high-frequency data.

The first comprehensive treatment of macroeconomic nowcasting was presented by Giannone, Reichlin, and Small (2008), who formalized a dynamic factor model (DFM) approach to track GDP in real time. Their framework, presented in the *Journal of Monetary Economics*, employed Kalman filtering techniques to handle the “jagged edge” of data availability—where diverse economic variables arrive with different publication delays and frequencies. By reducing large datasets into a smaller number of common factors, their model continuously updated GDP estimates with each data release, effectively quantifying how latest information (“news”) adjusts real-time GDP forecasts.

Building on this precise methodology, Giannone et al. (2016) released the New York Fed Staff Nowcast. The model will be in-depth analyzed in the next section.

Another important nowcasting tool is the GDPNow model, developed by Higgins (2014) at the Federal Reserve Bank of Atlanta. While this system builds heavily on the

foundations laid by Giannone, Reichlin, and Small (2008), it differs from the New York Fed Staff Nowcast in how it calculates GDP. Rather than providing a single aggregate forecast, GDPNow employs Bayesian Vector Autoregressions (BVAR) to generate forecasts for thirteen individual GDP subcomponents. These component predictions are then summed to form a real-time estimate of total GDP growth.

Together, both models serve as crucial tools for the Federal Reserve and, more broadly, for financial institutions, offering weekly updated forecasts on the state of the U.S. economy.

2.2. The New York Fed Staff Nowcast

The core of the New York Fed Staff Nowcast is its dynamic factor model (DFM), which is designed to condense a large number of economic time series into a small set of latent factors. These factors are assumed to capture the underlying, common dynamics of the economy, which are then used to update the GDP estimate. The model allows for real-time tracking of GDP by continuously integrating new data releases as they become available, adjusting the GDP forecast based on the discrepancies between the predicted and actual values of incoming indicators.

The use of latent factors is crucial, as it enables significant dimensionality reduction by summarizing information from diverse and numerous economic indicators, while smoothing out idiosyncratic noise. The model condenses this information into distinct blocks by assigning every series to a single “Global” factor that captures broad common economic dynamics. To better account for unique correlations within specific subgroups, the model also includes additional local blocks. The “Soft” block models survey-based data reflecting the perceptions and sentiments of economic agents, while the “Real” and “Labor” blocks control for correlations among real activity indicators and labor market variables, respectively. This

hierarchical structure enhances the model's interpretability and robustness by capturing both overarching economic trends and subgroup-specific idiosyncrasies.

Financial variables are not included in the model. Although these variables offer timely data, they are often volatile and have a limited impact on nowcasting GDP growth once a broad range of macroeconomic indicators is incorporated (Banbura et al. 2012). Nonetheless, financial conditions remain useful for assessing downside risks in the economy (Adrian et al. 2016).

The DFM is represented by the following equation:

$$y_t = \mu + \Lambda f_t + e_t \quad (2)$$

where:

- y_t is a vector of observed economic indicators at time t ,
- μ is a constant,
- Λ is the matrix of factor loadings,
- f_t is the vector of latent factors,
- e_t is the vector of errors or noise terms.

The model assumes that the observed data y_t can be explained by a small number of latent factors f_t , which evolve over time and are updated with each new data release.

To track these factors over time, the model employs Kalman filtering, which allows the system to update the estimate of the latent factors and the GDP forecast as new data points are observed. The Kalman filter operates iteratively, adjusting the state of the model by incorporating new information while minimizing prediction errors.

The update process in the Kalman filter is represented by the following recursive equations:

$$\hat{f}_{t|t} = \hat{f}_{t|t-1} + K_t(y_t - \hat{y}_{t|t-1}) \quad (3)$$

$$\hat{y}_{t|t} = \Lambda \hat{f}_{t|t} \quad (4)$$

where:

- $\hat{f}_{t|t-1}$ is the predicted value of the latent factors for time t , made at time $t - 1$,
- K_t is the Kalman gain matrix,
- y_t is the observed data at time t ,
- $\hat{y}_{t|t-1}$ is the predicted observation for time t , made at time $t - 1$.

The Kalman filter adjusts the latent factors $\hat{f}_{t|t}$ in real time as new economic data is released, updating the GDP forecast accordingly¹.

The New York Fed Staff Nowcast model is updated daily at 10 a.m. to incorporate each new data release, and the updated nowcast is published every Friday at 11:15 a.m. on the New York Fed's public website. This frequent updating process allows the model to integrate real-time data flows efficiently and measure the impact of individual data releases. Additionally, to ensure that the model remains responsive to evolving economic conditions, its parameters are re-estimated quarterly using the most recent 15 years of data.

2.3. Staff Nowcast 2.0

In late 2021, the New York Fed suspended the Staff Nowcast releases due to extreme volatility and data inconsistencies caused by the COVID-19 pandemic. However, in 2023, a revamped version of the model, referred to as the Staff Nowcast 2.0, was reintroduced. This new iteration introduces several innovative features designed to better address challenges posed by highly volatile and irregular economic data observed during the pandemic.

¹ The concepts of latent factors and Kalman gain are explained in detail in Chapter 4.2 (Nowcasting Procedure)

The new model is represented by the following equation:

$$y_t = \mu + \iota g_t + \Lambda f_t + e_t \quad (5)$$

with g_t calculated as follows:

$$g_t = g_{t-1} + \gamma_g v_{gt}, \quad v_{gt} \sim \text{iid } N(0,1) \quad (6)$$

where g_t represents the long-run growth trend, γ_g is the parameter governing the evolution of the trend, and v_{gt} is a normally distributed error term².

The inclusion of the long-run growth trend factor g_t represents a major update to capture the gradual decline in the average growth rate of U.S. real GDP, particularly noticeable since the early 2000s. This factor allows the model to account for shifts in long-term economic trends, enabling a better understanding of the economy's evolving trajectory.

Another key change in this version is the adoption of a Bayesian estimation approach, which not only generates point estimates of GDP growth but also provides probability intervals, enhancing the model's ability to handle uncertainty. This allows for more robust forecasting by capturing the variability in the data.

To tackle challenges presented by data volatility during the pandemic, the model incorporates time-varying volatility and outlier adjustment mechanisms for the latent factors. These updates allow the model to adapt to fluctuations in volatility and handle extreme economic shocks more effectively. Specifically, the model introduces time-varying volatility σ_{ft} and outliers s_{ft} to scale and adjust for large, unexpected shocks, particularly those associated with the COVID-19 crisis. These new dynamics allow for more precise modeling of

² In formula (5), ι is a selection vector that maps the trend g_t exclusively to GDP and income variables, giving a value of 1 to these series and 0 to the others.

the interactions between different economic variables, especially in the face of unprecedented economic disruptions.

Finally, the latent factor structure of the model has also been revised. As in the legacy version, all series load on a single Global factor. The "Soft" factor, which captures correlations in survey data and economic sentiment, and the "Labor" factor, which represents variables related to the labour market, remain unchanged. However, several additional factors have been introduced to capture other specific aspects of the economy. The "Nominal" factor replaces the previous "Real" factor, focusing specifically on price-related data or series expressed in nominal terms. Furthermore, a new "COVID" factor has been introduced to account for the economic disruptions caused by the COVID-19 pandemic. Active only from March to September 2020, this factor allows the model to adjust for the significant shocks during this period. To manage these disruptions, the model restricts the volatility of the COVID factor by setting $\sigma_{COVID,t} = 1$ and uses an outlier vector $s_{COVID,t}$ to appropriately scale these shocks.

2.4. Fiscal Variables in Nowcasting

Two recent studies highlight the potential, albeit limited, use of fiscal data in nowcasting macroeconomic variables. The first, by Róbert Ambrisko (2022), explores the use of high-frequency fiscal data, such as monthly social contributions and VAT receipts, for nowcasting key macroeconomic indicators in the Czech Republic. Ambrisko employs bridge-equation models and MIDAS regressions to link fiscal variables to macro variables like the wage bill and nominal GDP. His findings indicate that using monthly social contribution data leads to a reduction in the RMSE (Root Mean Square Error) of wage bill forecasts by about 2%, and incorporating daily fiscal data results in a 4% improvement. However, when using VAT data for nominal GDP nowcasting, the improvements are more modest, and the results are not consistently superior to institutional forecasts.

The second study, conducted by Alifatussaadah et al. (2019), investigates the role of fiscal variables in nowcasting Indonesia's quarterly GDP growth. By employing a dynamic factor model with a large dataset of 61 non-fiscal variables, the authors discover that although fiscal variables show a low correlation with GDP, their inclusion results in a slight improvement in early GDP growth forecasts. Specifically, incorporating fiscal data leads to a small reduction in the RMSE of these preliminary estimates.

Overall, these studies demonstrate that the inclusion of fiscal variables can improve nowcasting models, although the contribution remains modest.

3. Data

3.1. Data Sources and Vintages Construction

The empirical analysis is entirely based on real-time data and replicates the operational framework of the New York Fed Staff Nowcast. To reproduce genuine real-time forecasting conditions, a complete system of weekly vintages was developed, with each vintage reflecting exactly the information that would have been available on a specific Friday. In practical terms, this means that every dataset captures the values published up to that date, the revisions known at that time, and the actual release schedule of each indicator. For all variables, the historical sample extends back to January 1985, or to the earliest date of availability for series introduced later. A long history is needed for the Dynamic Factor Model, which requires enough data to estimate the latent factors and their loadings reliably. When a variable is created more recently, its sample naturally begins at the first public release, but it is still incorporated consistently into the model.

Two parallel sets of vintages were constructed. The first includes only the macroeconomic indicators used in the baseline version of the model, while the second adds the

fiscal variables introduced in this thesis to the same macroeconomic set. The weekly vintages cover the period between the first Friday of December 2016 and the last Friday of July 2025: the initial date marks the beginning of the forecasting cycle for the first quarter of 2017, and the final date corresponds to the last week used to forecast the second quarter of 2025. Consequently, the nowcasting window spans Q1 2017 to Q2 2025, for a total of thirty-four consecutive quarters.

All vintage series were built using ALFRED, the archival database of the Federal Reserve Bank of St. Louis, which provides every historical release of each variable exactly as originally published. Because ALFRED preserves the full revision history and the original publication dates, it is ideally suited for constructing real-time datasets consistent with the timing requirements of a nowcasting model.

3.2. Selection of Macroeconomic Variables

The set of macroeconomic variables used in the constructed models is a hybrid between the series included in the original Staff Nowcast and a restricted subset of those added in Staff Nowcast 2.0. The inclusion of only part of the expanded 2.0 dataset is not due to theoretical reasons but to practical constraints related to data availability, as some of the indicators introduced in the updated model are proprietary or lack vintage history. For this reason, the final dataset preserves the macroeconomic backbone of the original model and expands it only where complete, consistent, and public weekly vintages were available. A detailed comparison of the variable sets is provided in the Appendix: Table B1 lists the indicators included in the original Staff Nowcast, Table B2 reports those introduced in Staff Nowcast 2.0, and Table B3 presents the final selection of variables used in the models developed for this thesis.

3.3. Fiscal Variables

To evaluate whether fiscal information can improve GDP nowcasting, three fiscal indicators were incorporated into the model. These variables, listed in Appendix Table B4, were retrieved in real-time format from ALFRED and transformed consistently with the same procedures applied to the macroeconomic variables.

The selection of these variables was guided by their ability to capture distinct but complementary dimensions of fiscal activity that are relevant for economic fluctuations. The variable GCEC1 was chosen because it closely reflects the government expenditure component of national accounts. As a direct measure of federal consumption and investment, it offers a precise proxy for public spending and therefore captures the most immediate fiscal channel influencing aggregate demand.

The second indicator, W875RX1, measures federal government current receipts, primarily reflecting tax revenues. This series exhibits a pronounced procyclical behavior and is released at a monthly frequency, offering a relatively timely signal about the state of the economy. Because tax receipts tend to rise when economic activity strengthens and fall during downturns, their inclusion can help detect early signs of deceleration or acceleration in GDP growth, thereby enhancing the responsiveness of the nowcast.

Finally, MTSDS133FMS, which captures federal net lending/borrowing, was selected as an indicator of the fiscal balance. Periods of rising deficits are often associated with expansionary fiscal actions—such as increased public spending, tax cuts, or countercyclical interventions—that tend to support aggregate demand during downturns. This variable therefore provides a useful signal of the fiscal stance and the degree of policy accommodation, offering additional information that may help the model track short-run movements in GDP, particularly when government intervention becomes more prominent.

Overall, these three variables were selected not only because they are theoretically grounded and relevant, but also because they display a sufficiently strong cyclical relationship with economic activity to be informative for nowcasting. Together, they offer a balanced representation of government spending, government income, and the fiscal stance, allowing the model to incorporate a richer picture of the public sector’s contribution to economic growth.

4. Methodology

4.1 Model Specification

The empirical analysis builds upon the original version of the New York Fed Staff Nowcast, using the publicly available Python implementation released by the Federal Reserve Bank of New York³. This framework was selected primarily for computational reasons. Staff Nowcast 2.0 is available only in MATLAB, whereas the original model has a fully open and well-documented Python translation of the code⁴.

The core structure common to both versions is a Bayesian dynamic factor model composed of a global factor that loads on all variables, block-specific factors that capture sectoral dynamics, and a measurement equation in which all series enter in stationary form. Because the mechanism through which new indicators are integrated remains essentially unchanged, adopting the original version does not compromise the validity of evaluating the incremental informational content added by fiscal variables.

4.2. Nowcasting Procedure

³ The official repository of the New York Fed Staff Nowcast is available at: <https://github.com/FRBNY-TimeSeriesAnalysis/Nowcasting>

⁴ The Python translation is available at: <https://github.com/MajesticKhan/Nowcasting-Python>

The methodological framework adopted in this thesis is based on the construction of two structurally identical Dynamic Factor Models⁵. The first model, referred to as the baseline model, serves as the reference specification and incorporates only the macroeconomic indicators listed in Table B3. The second model, the fiscal-enhanced model, extends this framework by adding the fiscal variables presented in Table B4. Except for the presence or absence of these fiscal indicators, the two models are identical in every aspect: they follow the same factor structure, use the same data transformations, apply the same priors, and are estimated according to the same sequence of operations. This strict symmetry ensures that any divergence in forecasting performance can be attributed exclusively to the added fiscal variables.

The weekly nowcasting procedure replicates the operational logic of the New York Fed Staff Nowcast. Each forecasting cycle spans 22 weeks, corresponding to the twenty-two Fridays preceding the Bureau of Economic Analysis' advance release of quarterly GDP. For every weekly vintage within this window, the model incorporates only the information available up to that date. The Kalman filter updates the latent state based on newly released data or revisions, and the nowcast for the corresponding quarter is produced using this contemporaneous information set. This process ensures that the entire forecasting sequence is conducted without relying on ex-post revised data and without incurring look-ahead bias.

During the final month of quarter $t - 1$, the model generates the initial nowcast for the upcoming quarter t . Since no data for the new quarter is yet available, this forecast relies entirely on the historical dynamics of the latent factors. To formalize this, the model employs a state-space representation, a mathematical framework that links the unobserved state of the economy

⁵ The code developed for this thesis is available in the public GitHub repository: <https://github.com/mattecasta17/GDPnowcast-fiscal>

to the observed data. This structure consists of two key equations. First, a transition equation that describes how the hidden factors f_s evolve over time:

$$f_s = Af_{s-1} + Hu_s, \quad u_s \sim \mathcal{N}(0,1) \quad (7)$$

where A captures the autoregressive structure of the latent factors f_{s-1} , H governs the factor innovations (shocks), and u_s is the vector of structural innovations to the factors, assumed to be normally distributed.

Second, an observation equation that connects these hidden factors to the economic indicators:

$$x_s = \Lambda f_s + e_s \quad (8)$$

where x_s denotes the vector of all observed indicators at time s (including the target GDP variable y), Λ maps the factors into the observed indicators, and e_s denotes idiosyncratic disturbances. Building on this framework, the estimation process is initiated by using all information available up to the end of quarter $t - 1$, to obtain the Kalman smoother estimate of the latent factors at that date, $\hat{f}_{t|t-1}$. These factors are then projected forward through the transition equation (7), yielding the model-implied expectation for the beginning of quarter t :

$$\hat{f}_{t|t-1} = A\hat{f}_{t-1|t-1} \quad (9)$$

Finally, the projected factors are converted into a GDP forecast consistent with the observation equation (8). This is achieved by isolating the specific row of the loading matrix Λ corresponding to GDP—denoted here as λ'_y —and applying it to the factors:

$$\hat{y}_t^{(0)} = \lambda'_y \hat{f}_{t|t-1} \quad (10)$$

where $\hat{y}_t^{(0)}$ represents the baseline GDP projection before the arrival of any data releases.

Once the calendar quarter begins, the parameters of the Dynamic Factor Model are re-estimated using the data vintage corresponding to the first business day of the new quarter⁶. This quarterly re-estimation, consistent with the procedure adopted by the New York Fed, prevents the model from exploiting information not available in real time and ensures that the dynamics of the system can adjust to evolving macroeconomic relationships. After re-estimation, these parameters remain fixed throughout the quarter.

Each weekly data vintage updates the underlying information set used by the Kalman filter. Let X^{old} denote the dataset associated with the previous vintage and X^{new} the dataset associated with the current one. The model first computes the GDP forecast conditional on X^{old} , denoted y^{old} , and then recomputes the forecast incorporating the updated dataset, yielding y^{new} . The difference between these two quantities,

$$\Delta \hat{y}_{t|v} = \hat{y}_{t|v} - \hat{y}_{t|v-1} \quad (11)$$

represents the total weekly revision to the GDP nowcast.

To interpret this revision in terms of incoming information, each indicator $x_{j,s}$ is compared with its value in the previous vintage, producing the innovation (“news”)

$$news_{j,s} = x_{j,s}^{new} - x_{j,s}^{old} \quad (12)$$

which is nonzero only when the variable has been released or revised. In the dynamic factor structure, the impact of this innovation on the nowcast is determined by the Kalman gain⁷ and the propagation of factor dynamics to the target quarter. The contribution of indicator j associated with period s is therefore

$$impact_{j,s} = \lambda'_y A^{h(s)} K_{j,s} news_{j,s} \quad (13)$$

⁶ The Fed Staff Nowcast re-estimates the parameters using the most recent 15 years of available data, whereas the baseline and the fiscal-enhanced model use data from 2000 onward.

⁷ The Kalman gain is computed based on the estimated variance of the factors and the idiosyncratic errors, minimizing the mean squared error of the update.

where $K_{j,s}$ is the Kalman gain associated with that observation and $A^{h(s)}$ maps its effect to the quarterly horizon, with $h(s)$ representing the temporal distance to the target quarter. By construction, all contributions add up exactly to the total nowcast revision:

$$\hat{y}_{t|v} - \hat{y}_{t|v-1} = \sum_j \sum_s \text{impact}_{j,s} \quad (14)$$

This identity guarantees that every weekly revision of the GDP forecast is fully accounted for by economically interpretable news terms, thereby allowing a precise assessment of how each variable contribute to the evolution of the nowcast.

4.3. Accuracy Metrics for Model Comparison

To evaluate forecasting accuracy, a range of descriptive and diagnostic statistics was produced over the nowcast window covering Q1 2017 to Q2 2025. However, the comparison between the two model specifications is based on the three standard metrics most widely used in the nowcasting literature: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

RMSE places a larger penalty on big forecast errors because it squares the deviations before averaging them; as a result, it is particularly sensitive to occasional but sizable errors in the nowcast. It is calculated as follows:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2} \quad (15)$$

where $\hat{y}_t$ is the nowcast of GDP growth produced by the model for quarter t , while y_t is the BEA advance release of GDP growth for quarter t .

By contrast, MAE assigns an equal weight to all forecast errors. Its formula therefore is:

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |\hat{y}_t - y_t| \quad (16)$$

In addition to these accuracy measures, the Bias was also computed to assess whether either model tends to systematically overpredict or underpredict GDP growth across the nowcasting window:

$$\text{Bias} = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t) \quad (17)$$

These metrics are computed for each quarter and subsequently aggregated to assess whether the inclusion of fiscal variables improves predictive performance. The combination of parallel model construction, real-time data vintages, weekly 22-week forecasting windows, quarterly re-estimation, and standard accuracy metrics provides a rigorous and transparent methodology for examining the incremental informational value of fiscal indicators in a real-time nowcasting environment.

5. Empirical Results

5.1. Baseline Model Results⁸

As outlined in the previous section, RMSE and MAE serve as the primary metrics for evaluating and comparing the two models. Nonetheless, additional diagnostic measures were computed, and the most informative among them are presented in this chapter and contrasted across the two specifications. Throughout the analysis, the year 2020 is generally excluded from the calculations, as it represents an extreme outlier driven by the pandemic.

⁸ All metrics of the baseline model can be explored interactively in the accompanying dashboard, available at the following link: <https://gdpnowcast-fiscal-hfktqcuvctridprnu66rh3.streamlit.app/>. If the application is in sleep mode, the user is required to select 'Yes, get this app back up' and reload the page.

The evaluation of the baseline model shows stable performance in normal macroeconomic conditions and highlights the distortions introduced by the pandemic. Excluding 2020, the model achieves an average RMSE of 1.87 and an average MAE of 1.68. Including 2020 increases these metrics to 3.22 and 2.95, reflecting the extreme volatility of the pandemic period. As expected, 2020 exhibits by far the worst accuracy (RMSE of 14.18, MAE of 13.30), while 2021 and 2022 also show markedly higher errors compared with the other years. The elevated errors observed in 2022 and 2023 likely reflect the unusually volatile post-pandemic environment, characterised by rapid monetary tightening, large data revisions, and unstable trade and production dynamics, all of which made short-term economic conditions more difficult for the model to anticipate. A summary of the metrics across the period is shown in Figure A2.

The model does not display systematic bias overall. Excluding 2020, the average bias is essentially zero (0.04), with no persistent pattern across years. As illustrated in Figure A3, when breaking down the error in quarters Q1 shows the highest MAE (1.97), followed by Q2 (1.43), while Q3 and Q4 both average 1.26. A plausible explanation for the higher early-year errors is the well-documented residual seasonality embedded in early-year economic data (Owyang and Shell 2018). Including 2020 changes the ordering of the quarters, which is expected given that Q2 and Q3 were the most affected by the pandemic.

Analyzing the drivers behind the movements of the nowcast shows that updates are driven predominantly by new data releases, which account for roughly 80% of the total impact, while revisions represent about 20% and only become dominant in exceptionally volatile years.

When examining the average absolute impact by category, Survey indicators emerge as the most influential block, followed by Prices, Production, Labor, and Trade (Figure A4). A variable can generate a large impact for two reasons: either the model assigns a high loading to

that variable—reflecting its structural importance as a reliable indicator of economic conditions—or the forecast error (the “news”) is large because the variable is difficult to anticipate. Survey indicators fall into the latter category, as their forward-looking nature makes them prone to large deviations from model-based expectations. Prices, Production, and Labor instead tend to carry greater structural weight because, when released, they provide a clear assessment of current economic conditions. Trade shows a comparatively high absolute impact as well, likely reflecting the high volatility and noise typical of trade flows, which increases the difficulty of forecasting their short-term movements. Conversely, Consumption and Construction display more moderate absolute impacts, presumably because the model is better able to anticipate their trajectories. Interestingly, when the 2020 sample is included, Construction, Consumption, and Labor show a significant increase in their average impact. This suggests that during a major shock like the pandemic, these indicators become the most informative signals for economic activity.

This pattern is consistent with the interpretation advanced by Giannone, Reichlin and Small (2008), who show that survey measures tend to matter most at the start of the nowcasting cycle, when few hard indicators have yet been released and qualitative signals offer the earliest read on prevailing conditions. As progressively richer and more reliable data become available, the influence of surveys fades and the largest adjustments to the nowcast typically occur once the first hard data for the reference period are published.

Overall, the Baseline Model exhibits stable accuracy outside the pandemic years, shows almost no systematic bias, and relies on a coherent hierarchy of information, providing a robust benchmark against which to assess the added value of fiscal variables.

5.2. Fiscal-Enhanced Model Results⁹

The methodology of the fiscal-enhanced model mirrors the baseline specification, with the only difference being the inclusion of three fiscal indicators that enter the factor structure in the same way as the macroeconomic blocks.

Across the full sample, the fiscal-enhanced model achieves a systematic improvement in predictive accuracy relative to the baseline specification. Excluding 2020, the average RMSE declines from 1.87 to 1.81, while the MAE decreases from 1.68 to 1.63, indicating a consistent reduction in the magnitude of the errors. The bias is also reduced, falling from 0.04 to 0.01. Interestingly, the improvement is not limited to the overall averages: apart from 2020, the fiscal-enhanced model records lower RMSE and MAE in every single year of the sample. Even when 2020 is included, the fiscal specification still delivers a lower average RMSE and MAE than the baseline model. The same pattern does not hold for the bias. A summary of the accuracy metrics for the fiscal-enhanced model is presented in Figure A5.

A clear improvement also emerges when the analysis is broken down by quarter. As shown in Figure A6, the fiscal-enhanced specification reduces the MAE across all quarters: the Q1 error decreases from 1.97 to 1.94, while both Q3 and Q4 fall from 1.26 to 1.24. The largest improvement is observed in Q2, where the MAE drops from 1.43 to 1.31, amounting to a significantly greater reduction relative to the other quarters.

To further investigate this pattern, it is useful to look at the variable-level impacts of the fiscal indicators. A closer inspection shows that MTSDS133FMS (the Federal Surplus/Deficit) is by far the fiscal variable that contributes the most to the nowcast updates over the entire sample (Figure A7). Its average absolute impact is consistently higher than that of the other

⁹ All metrics of the fiscal-enhanced model can be explored interactively in the accompanying dashboard, available at the following link: <https://gdpnowcast-fiscal-4hjcd5clcthp5wh6rngws.streamlit.app/>. If the application is in sleep mode, the user is required to select 'Yes, get this app back up' and reload the page.

fiscal indicators and peaks in the second quarter, suggesting that this series carries disproportionately strong informational content for the DFM.

The behaviour of MTSDS133FMS is also consistent with the seasonality of U.S. federal cash flows, which typically display pronounced fluctuations during the spring months. The April tax season leads to a sharp inflow of personal income-tax payments, while May and June often feature substantial adjustments related to quarterly estimated tax payments and regular cyclical disbursements. These institutional dynamics tend to produce large month-to-month swings in the federal deficit, increasing both the magnitude and the volatility of MTSDS133FMS in Q2.

This pattern provides a plausible explanation for why the fiscal-enhanced model achieves its largest improvement in forecast accuracy in the second quarter; the strength of the fiscal signals is highest at a time when the baseline specification is typically more exposed to revisions in key macroeconomic indicators. In such periods, the additional information embedded in the fiscal series helps the DFM update the common factor more effectively, thereby reducing forecast errors.

Finally, it is worth noting that this improvement does not rely exclusively on MTSDS133FMS. In fact, a replicating model that excludes the other two fiscal variables achieves lower accuracy metrics. This is consistent with the theory of Dynamic Factor Models, according to which multiple indicators that share even a modest degree of co-movement with the business cycle can collectively enhance the precision of the estimated factor, since their idiosyncratic noise tends to average out in a large cross-section (Stock and Watson 2002).

This aggregate role of the fiscal block is best understood by looking at its cumulative impact over time, presented in Figure A8. In the pre-pandemic period (2017–2019), the cumulative impact of the fiscal category is small stable, always close to zero. This pattern

indicates that fiscal flows and deficits followed a predictable trajectory, one that the baseline DFM was able to anticipate with limited revisions. In other words, the low impact observed before the pandemic suggests that fiscal data did not deviate substantially from the model's expectations.

Starting in 2021, however, the picture changes dramatically. The cumulative fiscal impact rises sharply and remains elevated throughout the post-pandemic years. This behaviour aligns closely with macroeconomic developments: the expansion of federal deficits contributed to sizeable and frequent upward revisions of the nowcast. In this environment, fiscal indicators—especially the deficit measure MTSDS133FMS—carry significantly more informational weight, systematically correcting the common factor when newly released data point to stronger-than-expected fiscal activity.

Interestingly, despite this surge in fiscal-driven upward revisions, the average bias of the fiscal-enhanced model remains negative (around -0.2). This persistent underestimation implies that, even after incorporating the fiscal indicators, the model still tends to underestimate GDP growth. Such a result is consistent with the idea that the post-pandemic fiscal expansion had an even stronger positive effect on real activity than the model fully captures.

Overall, the sharp increase in cumulative fiscal impact—combined with the negative bias—supports the view that the rise in U.S. deficits since 2021 has played a substantial role in GDP growth.

5.3. Diebold–Mariano Test

To confirm the statistical relevance of the improvements obtained with the fiscal-enhanced model, a Diebold–Mariano (DM) test was performed on the differences in RMSE, MAE and

Bias between the fiscal specification and the baseline model¹⁰. The test was applied to four subsamples: (i) the full period, (ii) the sample excluding 2020, (iii) the pre-pandemic years (2017–2019) and (iv) the post-pandemic period (2021–2025).

The most informative results come from subsample (ii), where both RMSE and MAE improvements are highly significant, with very large DM statistics and p-values close to zero. This confirms that the inclusion of fiscal variables systematically reduces nowcast errors.

Comparing subsamples (iii) and (iv) reveals an interesting pattern. While the fiscal model already improves accuracy before the pandemic—again with highly significant accuracy statistics—the gains become substantially larger in the post-pandemic period, consistent with the previous obtained results. For Bias, the DM statistic in the pre-pandemic sample is negative and statistically significant, indicating that, unlike in other periods, the baseline model actually exhibited a significantly lower systematic error during the pre-pandemic period. A summary of the DM statistics and p-values are reported in Tables B5 and B6.

Overall, across all economically meaningful subsamples (excluding 2020, pre-2020, post-2020), the improvements in RMSE and MAE are statistically significant, providing convincing evidence that the fiscal model offers a robust and meaningful gain in predictive accuracy.

6. Conclusion

This thesis investigated whether incorporating fiscal variables improves the accuracy of U.S. GDP nowcasting. By reconstructing a strict real-time forecasting environment, the study evaluated the predictive value of federal fiscal indicators within the nowcasting framework employed by the Federal Reserve Bank of New York.

¹⁰ A detailed description of the DM test is provided in Appendix C.

The empirical evidence confirms that fiscal data carries valuable information for real-time economic nowcasting. The fiscal-enhanced model consistently reduces forecasting errors compared to the baseline macroeconomic specification. These gains are particularly strong in the second quarter, driven by the seasonality of tax receipts, and in the post-pandemic period, where large government deficits have played a central role in driving economic activity. Analysis at the variable level identifies the Federal Surplus/Deficit as the most influential indicator; however, the results show that relying on a combined set of fiscal variables is superior to using a single measure, as the combination helps isolate the true economic signal.

These findings have direct practical implications on GDP nowcasting. For central banks and financial institutions, integrating fiscal data into nowcasting models can reduce forecast errors. This is especially relevant during periods of unusual fiscal policy, where traditional macroeconomic variables may fail to fully capture the stimulus effects on aggregate demand.

Future research could expand on these findings by exploring higher-frequency fiscal data or more granular breakdowns of government spending. Additionally, estimating this model using more advanced frameworks—such as the updated Staff Nowcast 2.0, which accounts for time-varying volatility and outliers—could further refine the understanding of how fiscal dynamics interact with real-time macroeconomic forecasting.

To ensure transparency and facilitate reproducibility, the complete codebase developed for this study, along with the detailed results, has been made publicly available on a dedicated GitHub repository. A technical overview of the computational framework is provided in Appendix D.

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Appendices

Appendix A – Figures

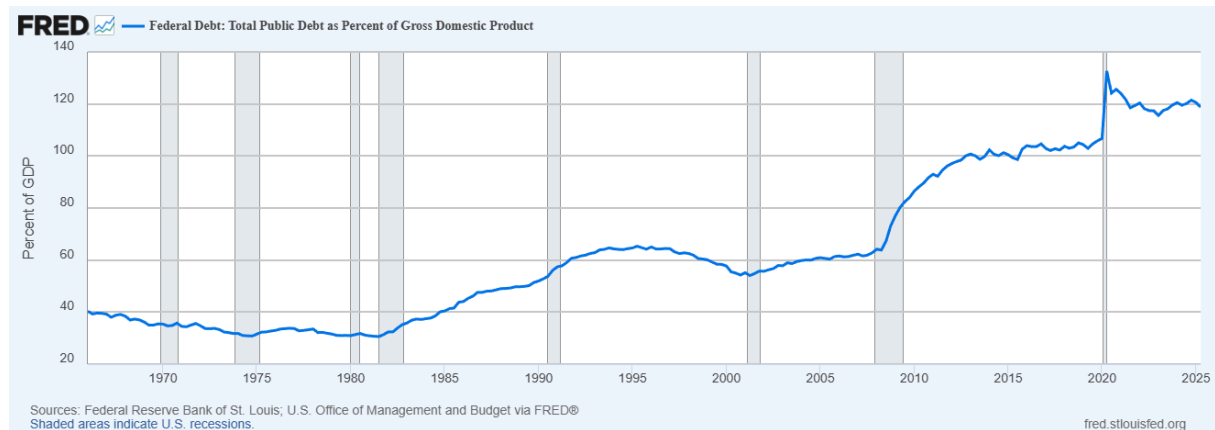


Figure A1: U.S. Total Public Debt (1966-2025)

The figure shows the evolution of the U.S. Debt-to-GDP ratio. From 2010 to the present, the ratio surged from approximately 60% to a peak of 140% in 2020, reflecting the massive fiscal expansion started after the 2008 financial crisis.

Source: Federal Reserve Economic Database (FRED)

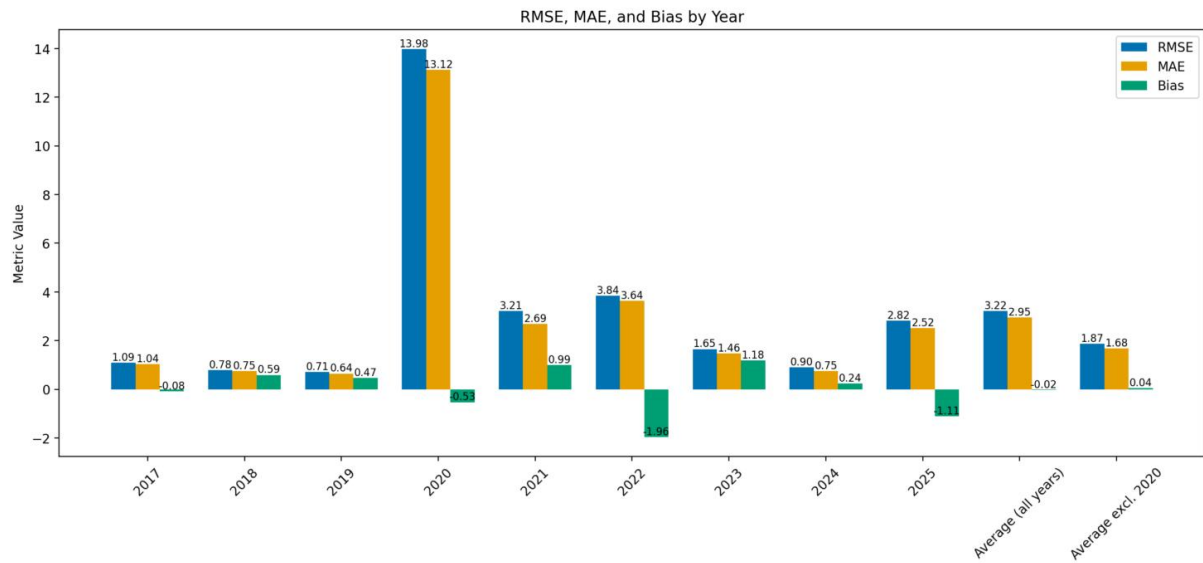


Figure A2: Accuracy metrics (2017-2025) – baseline model

The figure presents the accuracy metrics for the baseline model over the 2017-2025 period. Metrics remain relatively stable throughout the period, with the notable exception of 2020, where forecast errors spike significantly, highlighting the limitations of the standard specification in capturing extreme economic shocks. Average values are reported for both the full sample and the subsample excluding the 2020 outlier.

Source: Author calculations

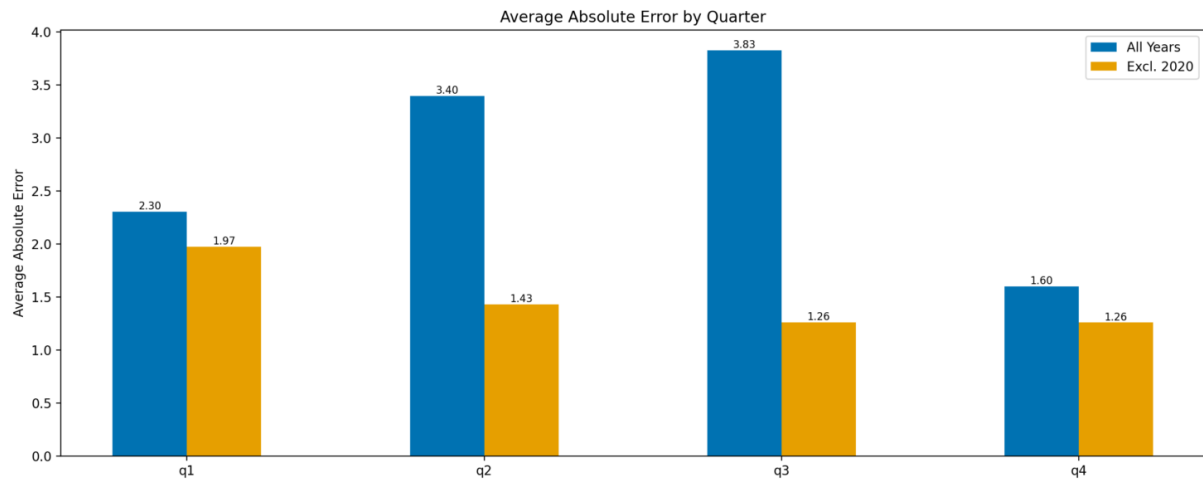


Figure A3: Average absolute error by quarter – baseline model

The figure breaks down the average absolute error (MAE) by quarter for the baseline model.

Excluding 2020, the highest errors occur in the first half of the year, likely due to residual seasonality.

When the full sample is considered, the ranking shifts, as Q2 and Q3 exhibit the largest deviations due to the severe impact of the COVID-19 shock.

Source: Author calculations

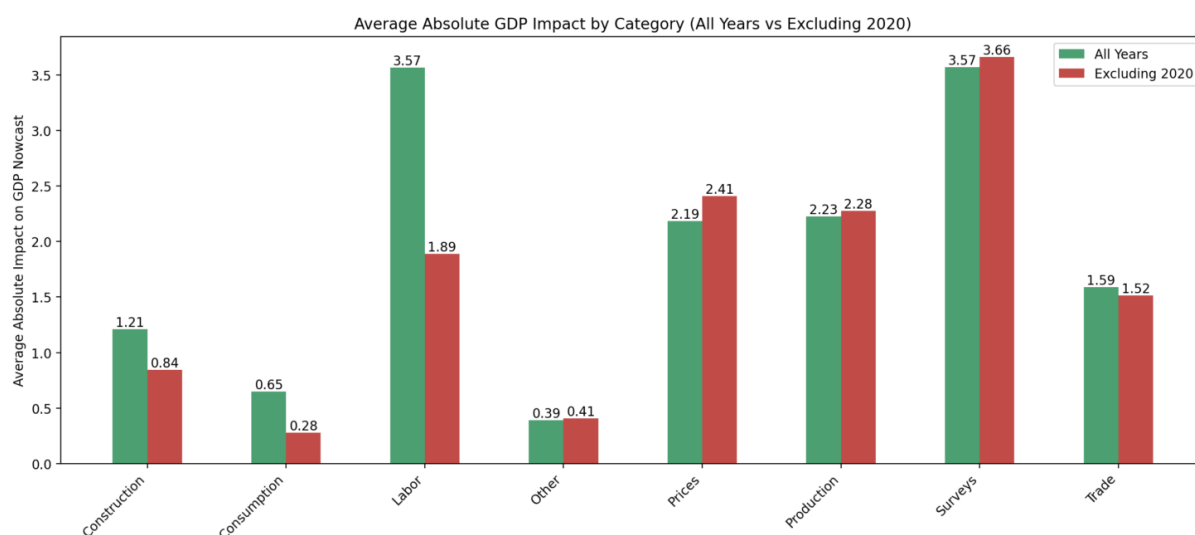


Figure A4: Average annual impact by category – baseline model

The figure ranks variables categories based on their average absolute impact on the GDP forecast.

Survey indicators emerge as the most influential block due to their high volatility and 'news' content, while Consumption and Construction exhibit more moderate impacts, inferring that the model anticipates their trajectories with greater accuracy.

When the 2020 sample is included, Construction, Consumption, and Labor show a significant increase in their average impact, suggesting that during a major shock like the pandemic, these indicators become the most informative signals for economic activity.

Source: Author calculations

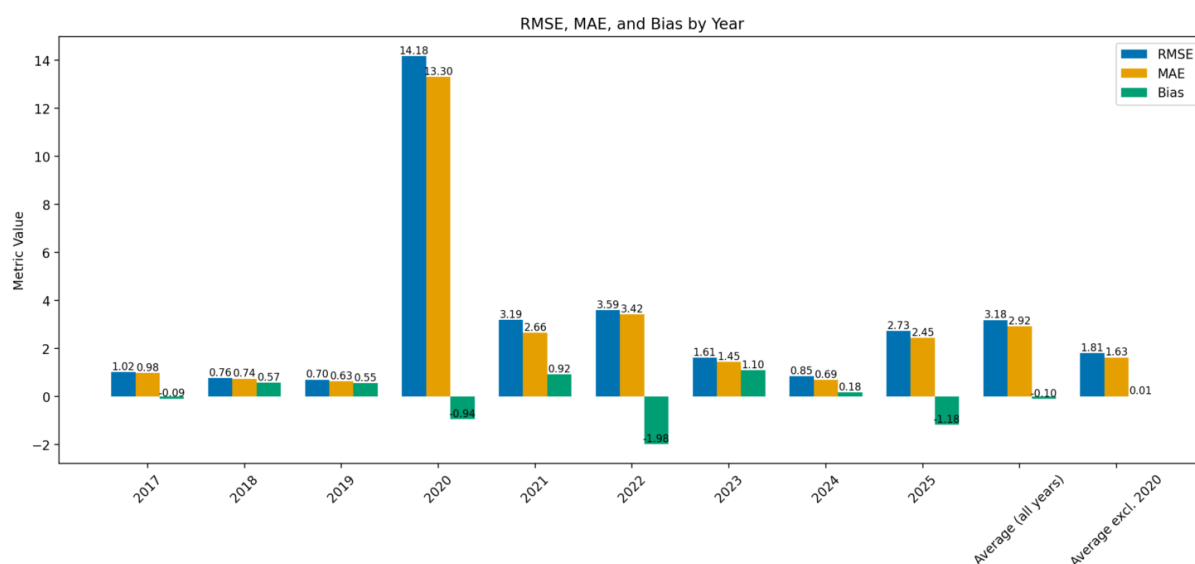


Figure A5: Accuracy metrics (2017-2025) – fiscal-enhanced model

The figure presents the accuracy metrics for the fiscal-enhanced model over the 2017-2025 period.

Notably, with the exception of 2020, predictive performance improves in every year. Excluding the 2020 outlier, the average RMSE decreases from 1.87 to 1.81, the MAE from 1.68 to 1.63, and the bias from 0.04 to 0.01, confirming the valuable contribution of fiscal variables to the GDP nowcast.

Source: Author calculations

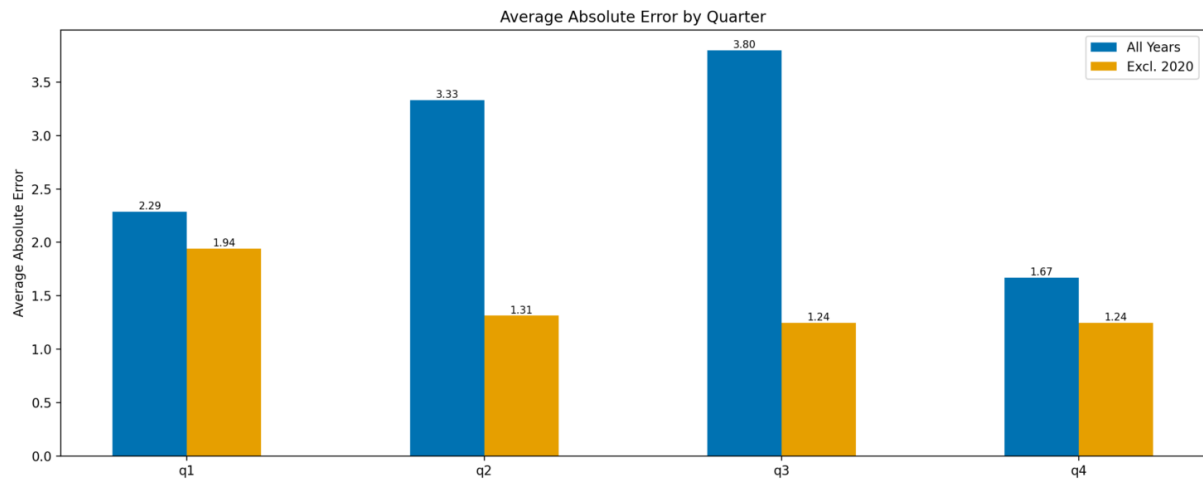


Figure A6: Average absolute error by quarter – fiscal-enhanced model

The figure displays the average absolute error (MAE) by quarter for the fiscal-enhanced model. While predictive accuracy improves across all quarters relative to the baseline, the most significant gain is observed in Q2, where the MAE drops from 1.43 to 1.31. This pronounced improvement likely stems from the model capturing the seasonality of tax receipts.

Source: Author calculations

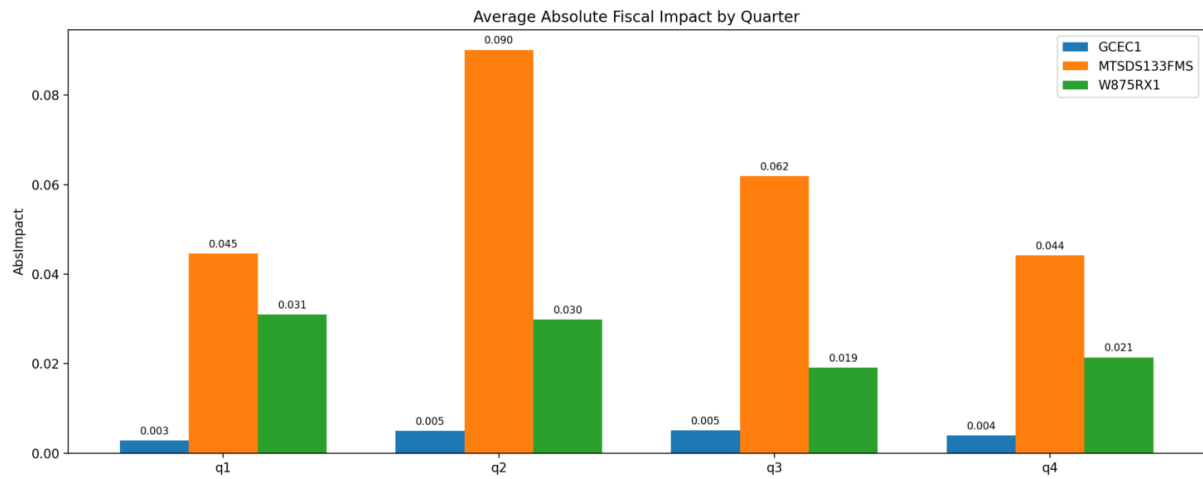


Figure A7: Average absolute impact of fiscal variables – fiscal-enhanced model

The figure illustrates the average absolute impact of fiscal variables on the GDP forecast, broken down by quarter. The strongest impact is observed in Q2. This aligns with the significant reduction in forecast error for the same quarter shown in the previous figure, hinting that the heightened informational content of fiscal variables during the tax season directly contributes to the improved accuracy of the nowcast.

Source: Author calculations

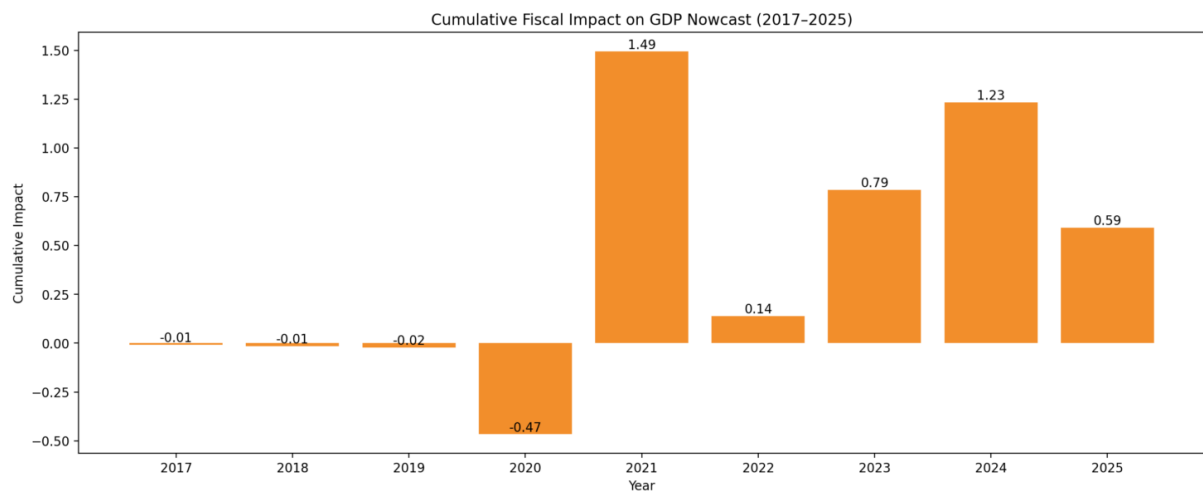


Figure A8: Cumulative impact of fiscal variables (2017-2025) – fiscal-enhanced model

The figure presents the cumulative impact of the fiscal block on the GDP nowcast over the 2017-2025 period. While the impact is negligible prior to 2020 due to predictable fiscal trends, it rises sharply from 2021 onward, reflecting how unprecedented fiscal expansion systematically drove upward revisions. Despite this contribution, the model’s persistent negative bias suggests that the actual boost to real activity from the fiscal stimulus was even stronger than what the model captures.

Source: Author calculations

Appendix B – Tables

Series ID	Series Name
PAYEMS	Payroll Employment
JTSJOL	Job Openings
GDPC1	Real Gross Domestic Product
CPIAUCSL	Consumer Price Index
DGORDER	Durable Goods Orders
HSN1F	New Home Sales
RSAFS	Retail Sales
UNRATE	Unemployment Rate
HOUST	Housing Starts
INDPRO	Industrial Production
PPIFIS	Producer Price Index
DSPIC96	Personal Income
BOPTXP	Exports
BOPTIMP	Imports
WHLSLRIMSA	Wholesale Inventories
TTLCONS	Construction Spending
IR	Import Price Index
CPILFESL	Core Consumer Price Index
PCEPILFE	Core PCE Price Index
PCEPI	PCE Price Index
PERMIT	Building Permits
TCU	Capacity Utilization Rate
BUSINV	Business Inventories
ULCNFB	Unit Labor Cost
IQ	Export Price Index
GACDISA066MSFRBNY	Empire State Mfg Index
PCEC96	Real Consumption Spending
A261RX1Q020SBEA	Real Gross Domestic Income
GACDFSA066MSFRBPHI	Philadelphia Fed Mfg Index

Table B1: Macroeconomic variables used in the original New York Fed Staff model

The table lists the macroeconomic variables included in the original specification of the New York Fed Staff Nowcast model.

Source: <https://github.com/FRBNY-TimeSeriesAnalysis/Nowcasting>

Series ID	Series Name
PAYEMS	Payroll Employment
JTSJOL	Job Openings
GDPC1	Real Gross Domestic Product
CPIAUCSL	Consumer Price Index
DGORDER	Durable Goods Orders
HSN1F	New Home Sales
RSAFS	Retail Sales
UNRATE	Unemployment Rate
HOUST	Housing Starts
INDPRO	Industrial Production
DSPIC96	Personal Income
BOPTEXP	Exports
BOPTIMP	Imports
WHLSLRIMSA	Wholesale Inventories
TTLCONS	Construction Spending
IR	Import Price Index
CPILFESL	Core Consumer Price Index
PCEPILFE	Core PCE Price Index
PCEPI	PCE Price Index
PERMIT	Building Permits
BUSINV	Business Inventories
IQ	Export Price Index
GACDISA066MSFRBNY	Empire State Mfg Index
PCEC96	Real Consumption Spending
A261RX1Q020SBEA	Real Gross Domestic Income
GACDFSA066MSFRBPHI	Philadelphia Fed Mfg Index
ADPMNUSNERSA	ADP nonfarm private payroll employment
AMDMVS	Manufacturer's shipments: Durable goods
AMDMUO	Manufacturers' unfilled orders: All industries
AMDMTI	Manufacturers' inventories: Durable goods
PRS85006112	Nonfarm business sector: Unit labor cost

Table B2: Macroeconomic variables used in the New York Fed Staff 2.0 model

The table presents the macroeconomic variables included in version 2.0 of the New York Fed Staff Nowcast model.

Source: <https://github.com/FRBNY-RG/New-York-Nowcast>

Series ID	Series Name
PAYEMS	Payroll Employment
JTSJOL	Job Openings
GDPC1	Real Gross Domestic Product
CPIAUCSL	Consumer Price Index
DGORDER	Durable Goods Orders
HSN1F	New Home Sales
RSAFS	Retail Sales
UNRATE	Unemployment Rate
HOUST	Housing Starts
INDPRO	Industrial Production
PPIFIS	Producer Price Index
DSPIC96	Personal Income
BOPTEXP	Exports
BOPTIMP	Imports
WHLSLRIMSA	Wholesale Inventories
TTLCONS	Construction Spending
IR	Import Price Index
CPILFESL	Core Consumer Price Index
PCEPILFE	Core PCE Price Index
PCEPI	PCE Price Index
PERMIT	Building Permits
TCU	Capacity Utilization Rate
BUSINV	Business Inventories
ULCNFB	Unit Labor Cost
IQ	Export Price Index
GACDISA066MSFRBNY	Empire State Mfg Index
PCEC96	Real Consumption Spending
A261RX1Q020SBEA	Real Gross Domestic Income
GACDFSA066MSFRBPHI	Philadelphia Fed Mfg Index
AMDMVS	Manufacturer's shipments: Durable goods
AMDMUO	Manufacturers' unfilled orders: All industries
AMDMTI	Manufacturers' inventories: Durable goods

Table B3: Macroeconomic variables used in the thesis models

The table displays the macroeconomic variables included in both the baseline and the fiscal-enhanced model created for the thesis.

Series ID	Series Name
GCEC1	Real Government Consumption and Gross Investment
W875RX1	Federal Government Current Receipts
MTSDS133FMS	Federal Net Lending/Borrowing

Table B4: Fiscal variables used in the fiscal-enhanced model

The table presents the fiscal variables incorporated into the fiscal-enhanced model. GCEC1 is observed at a quarterly frequency, while the other two are monthly series.

Period	Observations	RMSE	MAE	Bias
Full sample	652	-0.7659	1.8194	4.4093
Full excl. 2020	575	6.4012	6.4811	2.7804
Pre-2020	228	5.6154	3.8905	-2.9637
Post-2020	347	5.9777	5.6311	3.9278

Table B5: DM statistics for the fiscal block

This table reports the Diebold-Mariano test statistics across four distinct subsamples covering the 2017-2025 period. In every sample that excludes 2020, the fiscal-enhanced model yields statistically significant improvements in RMSE and MAE, as indicated by the large positive values. The only exception is the pre-2020 Bias, where the baseline specification exhibits a lower systematic error.

Source: Author calculations

Period	Observations	RMSE	MAE	Bias
Full sample	652	0.4444	0.0693	0.0000
Full excl. 2020	575	0.0000	0.0000	0.0056
Pre-2020	228	0.0000	0.0001	0.0034
Post-2020	347	0.0000	0.0000	0.0001

Table B6: P-values for the fiscal block

This table displays the p-values associated with the DM test statistics. Values close to zero indicate that the null hypothesis of equal predictive accuracy is rejected at standard significance levels. These results statistically confirm that, excluding 2020, the improvements in the accuracy metrics achieved by the fiscal-enhanced model are significant and not attributable to chance.

Source: Author calculations

Appendix C – The Diebold-Mariano Test

The Diebold-Mariano (DM) test, originally proposed by Diebold and Mariano (1995), represents the standard statistical procedure for comparing the predictive accuracy of two competing forecasting models. Its primary purpose is to determine whether the observed difference in loss functions—such as RMSE or MAE—is statistically significant or merely the result of random sampling variability. Formally, the test evaluates the null hypothesis of equal predictive accuracy, defined as $H_0: E[d_t] = 0$. The central component of the test is the loss differential series d_t , which is calculated as the difference between the loss of the baseline model and that of the fiscal-enhanced model:

$$d_t = L(e_t^{(baseline)}) - L(e_t^{(fiscal)}) \quad (C.1)$$

In this application, $L(\cdot)$ corresponds to squared errors (RMSE), absolute errors (MAE) and signed errors (Bias). The DM statistic is then computed as the t-ratio of the sample mean of this differential:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{Var}(\bar{d})}} \quad (C.2)$$

where $\bar{d}$ is the mean loss differential and $\widehat{Var}(\bar{d})$ is its HAC-consistent variance, which corrects for potential serial correlation and heteroskedasticity in the forecast errors. This adjustment is necessary to ensure valid inference in the test, since nowcast revisions typically exhibit temporal dependence. Under the null hypothesis, the test statistic asymptotically follows a standard normal distribution $N(0,1)$. Consequently, given the construction of the differential in equation (C.1), a statistically significant positive value implies that the loss of the baseline model is systematically larger than that of the fiscal model, thereby confirming the superior accuracy of the fiscal-enhanced specification.

To facilitate inference, p-values are reported alongside the test statistics. These values represent the probability of observing a test statistic at least as extreme as the one obtained, under the null hypothesis of equal predictive accuracy. A p-value below the standard significance thresholds (e.g., 0.05 or 0.01) provides strong evidence to reject the null hypothesis, confirming that the improvement in forecasting performance is statistically significant.

Appendix D – Code Repository

To ensure full transparency and facilitate the reproducibility of the empirical findings presented in this thesis, the complete codebase has been made publicly available on a dedicated [GitHub repository](#).

The computational core of this project is built upon a Python implementation of the New York Fed Staff Nowcast model, which has been adapted to incorporate the specific fiscal variables analyzed in this study and to simulate the strict real-time forecasting environment described in the methodology.

Two interactive dashboards were developed to allow for a granular exploration of the results. These tools enable users to visualize the real-time evolution of GDP forecasts and compare the [Baseline model](#) against the [Fiscal-Enhanced model](#). Each dashboard is structured around three levels of analysis: a granular single-quarter inspection, a multi-quarter overview of annual performance, and a comprehensive multi-year comparison.

The dashboard dedicated to the Fiscal-Enhanced model features an additional analytical layer at the end of the multi-year section, reporting the results of the Diebold-Mariano test to statistically assess the predictive contribution of the fiscal variables.

These applications are built using Streamlit—an open-source Python framework designed to create interactive data apps—and they enter sleep mode after 12 hours of inactivity; should this occur, clicking 'Yes, get this app back up' followed by a page reload is required to restore access.